

A Data-Driven Iterative Decoupling Feedforward Control Strategy With Application to an Ultraprecision Motion Stage

Yi Jiang, Yu Zhu, *Member, IEEE*, Kaiming Yang, Chuxiong Hu, *Member, IEEE*, and Dongdong Yu

Abstract—This paper develops a data-driven decoupling feedforward control scheme with iterative tuning to meet the challenge of the crosstalk problem in multiple-input multiple-output (MIMO) motion control systems. In contrast to model-based approaches, iterative tuning fully utilizes the available data to address the practical difficulty in obtaining an accurate dynamic model. The MIMO feedforward signal is iteratively updated by minimizing the developed crosstalk criterion. Specifically, to make the optimal problem convex, the MIMO feedforward controller is structuralized with a finite impulse response (FIR) filter and is parameterized by corresponding coefficients. A data-driven unbiased gradient approximation based on the Toeplitz matrix is then developed for updating the parameter vector. Furthermore, to deal with the Hessian inverse problem encountered in the numerical calculation of the update law, a stable inversion method combined with singular value decomposition is employed. The basic characteristics of the proposed scheme, including convergence accuracy and convergence rate, are illustrated through simulation. Finally, the proposed data-driven decoupling control scheme is applied to a developed ultraprecision motion stage, and the results show that the approach can significantly attenuate the servo error caused by the crosstalk problem. This simplicity and accuracy oriented control method without need of dynamic modeling is definitely suitable for industrial applications.

Index Terms—Crosstalk, data-driven tuning, decoupling feedforward control, gradient approximation, multiple-input multiple-output (MIMO) system.

I. INTRODUCTION

MODERN high-speed precision motion systems, such as numerical control machine tools, automatic inspection instruments, and semiconductor scanners, usually require excellent multiple-degree-of-freedom (DOF) motion control performance [1]–[3]. Generally, control architecture with feedback and feedforward is employed for these systems to guarantee robust stability and excellent tracking performance, respec-

tively [4]–[6]. Due to structure uncertainty and modeling error that existed in practical multiple-input multiple-output (MIMO) motion systems, internal coupling among DOFs inevitably appears and becomes one of the main obstacles for achieving excellent servo performance. This internal coupling characteristic is deemed as the crosstalk problem [7]. Therefore, various decoupling feedback methods are proposed to address the crosstalk problem [8]–[10]. Nevertheless, these existing decoupling feedback control schemes are usually too complex to realize in practical applications, and multi-intersected feedback paths also may render internal uncertainty. To avoid feedback loop redundancy and simplify the control system, a common control configuration consisting of single-input single-output (SISO) feedback control and decoupling feedforward control is developed [11], [12]. The SISO feedback part is to guarantee system robust stability under the diagonal dominance assumption, which can be fulfilled by the system initial dynamic decoupling in most MIMO systems. Moreover, the decoupling feedforward part is to attenuate the effect of the residual crosstalk for achieving excellent servo performance.

In general, as the feedforward control force accounts for the major control force, the feedforward control is crucial for the underlying tracking performance. To attenuate the effect of the crosstalk, the feedforward controller should compensate the plant model as accurately as possible. Many feedforward methods [13]–[15] have been proposed to obtain the inverse model, but the tracking performance of these model-based approaches is limited in practice due to the unavoidable model uncertainty, modeling error, and nonminimum phase zeros. To address this problem, an adaptive iterative control algorithm without demand for an accurate model is a good alternative. Typically, iterative learning control (ILC) [16]–[18] is an effective method that generates the control signal from the servo error information iteratively. However, the ILC technique is sensitive to set-point variations, thus limiting its application in occasions requiring variant trajectories.

Considering that model-based approaches are usually not easy to deal with the complex dynamic behavior in most industrial applications, another iterative control way with data-driven tuning is developed to obtain the control signal through system data collection [19]–[23]. In most cases, the controller is iteratively updated through a gradient-based algorithm to minimize an H_2 performance criterion. A typical data-driven tuning method is iterative feedback tuning (IFT) [24], which uses the control system itself to estimate the gradient. Due to the operational complexity and the low signal noise ratio, IFT is rarely applied to the decoupling feedforward controller

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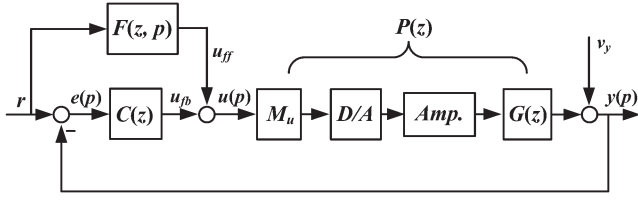


Fig. 1. Block diagram of a typical control configuration.

design. Another effective approach uses a perturbation method to estimate the gradient, but the large number of trials and the unknown method for choosing the perturbation limit its application [25].

To deal with the ubiquitous crosstalk problem in practical MIMO systems, a decoupling feedforward control scheme optimized by the measurement data is proposed in this paper. Specifically, the proposed scheme iteratively updates the control signal by minimizing the developed crosstalk criterion. To make the optimal problem convex, the controller is assumed to be linear and is then structuralized with the finite impulse response (FIR) filter. A data-driven unbiased gradient approximation based on the Toeplitz matrix is then developed, which remarkably simplifies the control scheme. Furthermore, a stable inversion method combined with singular value decomposition is employed to deal with the Hessian inverse problem. Herein, the determination of the controller structure, the acquisition of the gradient, and the handling of the Hessian inverse problem are the main focuses of the control scheme. In contrast to the traditional model-based approaches, the scheme automatically builds an effective feedforward decoupling controller based on the collecting data. Finally, significant attention is paid to the industrial implementation of the proposed data-driven control scheme to illustrate the practical performance.

II. DATA-DRIVEN ITERATIVE DECOUPLING FEEDFORWARD CONTROL STRATEGY

Fig. 1 shows the widely used multi-DOF control framework, wherein r , e , v_y , u , and y are the reference trajectory, servo error, immeasurable disturbance, plant input, and plant output, respectively. $G(z)$ denotes a MIMO linear time-invariant plant model (z is the forward time-shift operator). The feedback controller $C(z)$ is a diagonal matrix that consists of multiple SISO controllers. $F(z, p)$ is the MIMO feedforward controller that is parameterized by a parameter vector $p = [p_0 \ p_1 \ \dots \ p_r]^T$. The static input decoupling transformation matrix M_u is employed to achieve diagonal dominance, which depends on the computer-aided model. For simplicity, $P(z)$ represents the assembly of M_u , D/A , Amp , and $G(z)$.

To simplify the presentation of the data-driven decoupling strategy, a 2×2 diagonal dominant plant $P(z)$ is considered in this section. Fig. 2 depicts the configuration of the 2-DOF control system. The plant $P(z)$, the feedback controller $C(z)$, and the feedforward controller $F(z, p)$ are described as

$$P(z) = \begin{bmatrix} P_{xx}(z) & P_{xy}(z) \\ P_{yx}(z) & P_{yy}(z) \end{bmatrix} \quad C(z) = \begin{bmatrix} C_x(z) & 0 \\ 0 & C_y(z) \end{bmatrix} \\ F(z, p) = \begin{bmatrix} F_{xx}(z, p) & F_{xy}(z, p) \\ F_{yx}(z, p) & F_{yy}(z, p) \end{bmatrix}. \quad (1)$$

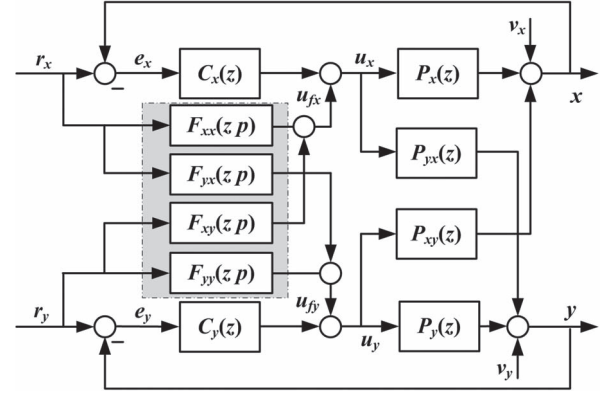


Fig. 2. Control configuration of a 2-DOF system.

In this paper, the feedforward controller is assumed to be linear since the characteristic is one of essential conditions for making the optimal problem convex, which will be discussed in Section II-A. Thus, the feedforward controller can be defined as

$$F_{ij}(z, p) = p_{ij}^T \Gamma(z), \quad i, j \in x \text{ or } y \quad (2)$$

where $\Gamma(z)$ is the basis function of the feedforward controller, and the parameter vector p is constructed by p_{xx} , p_{xy} , p_{yx} , and p_{yy} . The feedforward control design will be completed mainly by the determinations of three parts: 1) the update law of the parameter vector p ; 2) the basis function $\Gamma(z)$ for the controller structure design; and 3) the method of obtaining the gradient approximation based on the update law and the structure. These parts will be presented in the following sections.

A. Update Law of the Parameter Vector p

The data-driven MIMO feedforward controller is optimized by a gradient-based algorithm. The motivation of the tuning is to improve the servo performance by minimizing the objective function. Considering that the servo error is the crucial performance index, the objective criterion is chosen as

$$J(p) = e(p)^T \lambda e(p) \quad (3)$$

where $e(p) = [e_x^T(p) \ e_y^T(p)]^T$ denotes the servo error vector, and λ is a diagonal weight matrix. The algorithm is to obtain an optimal parameter vector p_d that satisfies

$$p_d := \arg \min_p J(p). \quad (4)$$

The value of parameter vector p iteratively updates by a gradient search, and the updates law can be expressed as [24]

$$p^{k+1} = p^k - \gamma^k R^{-1} \nabla J|_{p^k} \quad (5)$$

where k is the number of the iterative trials, R is a certain positive definite matrix, and the positive real scalar γ denotes the step size. The gradient can be derived from the objective function as

$$\nabla J|_{p^k} = 2 \nabla e(p^k)^T \lambda e(p^k) = 2 \left[\frac{\partial e^T(p)}{\partial p_0} \dots \frac{\partial e^T(p)}{\partial p_r} \right] \lambda e(p). \quad (6)$$

Generally, the Newton method is widely used to build the update law, which defines R in (5) as being equal to the Hessian matrix. For applications, the Gauss–Newton direction is a common choice to estimate the Hessian matrix due to its high convergence rate and accuracy [24]. The Hessian matrix is subsequently approximated as

$$R = \nabla^2 J|_{p^k} \approx 2\nabla e^T(p^k) \lambda \nabla e(p^k) \quad (7)$$

which leads to the determination of R .

To achieve the convergence of the iteration, the optimal problem needs to be convex [26]. Thus, the optimal problem could be convex by satisfying two assumptions: 1) The objective function $J(p)$ is convex; 2) the servo error $e(p)$ is an affine space with regard to parameter vector p . The first assumption is fulfilled when the Hessian $\nabla^2 J(p)$ is positive definite. By the approximation in (7), the Hessian is positive definite if and only if λ is positive definite and $\nabla e(p) \neq 0$. The second assumption is determined by the controller structure synthesis, which will be discussed in Section II-B.

B. Synthesis of Controller Structure and Determination of $\Gamma(z)$

According to (2), the structure of the feedforward controller is determined by the basis function $\Gamma(z)$. This research employs the FIR filter structure to construct the basis function, whose coefficients are replaced with parameter vector p . There are several issues considered for choosing the controller. First, the FIR filter is a polynomial that is able to approximate the inverse model. Second, the linear characteristic of the controller makes the overall servo error as an affine space with regard to parameter vector p , and the affine transformation is

$$e(p) = -\frac{Pr\Gamma^T}{1+PC}p + \frac{1}{1+PC}(r - v_y) \quad (8)$$

which satisfies the second assumption listed in Section II-A.

Furthermore, as the nonminimum phase characteristic will render the response lagging, which is unavoidable in practical systems, the noncausal characteristic of $\Gamma(z)$ is developed as

$$\Gamma(z) = \left\{ \underbrace{\dots z^2, z^1}_{s_1}, 1, \underbrace{z^{-1}, z^{-2}, z^{-3}, \dots}_{s_2} \right\} \quad (9)$$

where s_1 is the ahead order to compensate the noncausal behavior, and s_2 is the lagging order to fit the plant dynamics, which are determined by the time delays and the system resonances, respectively. In addition, the structure has a benefit of obtaining the gradient. Once the term $\nabla e(p_o)$ is obtained, the whole gradient $\nabla e(p)$ can be immediately calculated since the term $\nabla e(p_i)$ is equal to the term $\nabla e(p_o)$ by a delay of i sampling times, which $\nabla e(p_i) = \nabla e(p_o)z^{-i}$. This feature simplifies the practical implementation of the algorithm.

C. Approximation Law for Gradient $\nabla e(p)$

Here, we present the determination method for gradient $\nabla e(p)$. A discrete time-invariant system can be described in

state space as

$$\begin{cases} x(j+1) = Ax(j) + Bu(j) \\ y(j) = Cx(j) + Du(j) \end{cases}, \quad j = 0, 1, \dots, n-1 \quad (10)$$

where $x(j)$, $u(j)$, and $y(j)$ denote the state variable and the system input and output, respectively. Moreover, A , B , C , and D correspond to state-, control-, output state-, and output control-coefficient matrices, respectively. The output $y(j)$ can be defined as

$$\begin{bmatrix} y(0) \\ y(1) \\ \vdots \\ y(n-1) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} x(0) + \underbrace{\begin{bmatrix} D & 0 & \dots & 0 \\ CB & D & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ CA^{n-2}B & \dots & CB & D \end{bmatrix}}_{\Omega} \begin{bmatrix} u(0) \\ u(1) \\ \vdots \\ u(n-1) \end{bmatrix} \quad (11)$$

where the term Ω represents the Toeplitz matrix, whereas the Markov parameters $\Omega_{j,1}$ represent the system output at the j th sample time when a unit impulse is injected. The system output in Fig. 1 can be described by the Toeplitz matrix with inputs as

$$y(p) = T_r r + S v_y + T u_{ff}(p) \quad (12)$$

where T_r , S , and T represent the complementary sensitivity Toeplitz matrix, the sensitivity Toeplitz matrix, and the process sensitivity matrix, respectively. For the 2-DOF system mentioned in Fig. 2, the servo error $e(p)$ can be described as

$$\begin{aligned} e_x(p) &= S_{xx}(r_x - v_x) + S_{xy}(r_y - v_y) + T_{xx}u_{fx}(p) + T_{xy}u_{fy}(p) \\ e_y(p) &= S_{yx}(r_x - v_x) + S_{yy}(r_y - v_y) + T_{yx}u_{fx}(p) + T_{yy}u_{fy}(p) \end{aligned} \quad (13)$$

where the sensitivity Toeplitz matrix $S_{ij}(i, j \in x, y)$ is the map between the j -direction reference input and the i -direction error e_i . The process sensitivity Toeplitz matrix T_{ij} is the map between the j -direction process input u_{fj} and the i -direction servo error e_i . The process input u_{fi} includes the contributions of each direction reference trajectory, and it can be defined as

$$\begin{aligned} u_{fi} &= [F_{ii} \ F_{ij}] \begin{bmatrix} r_i \\ r_j \end{bmatrix} \\ &= \left[\sum_{m=-s_1}^{s_2} p_{ii}(m)z^{-m} \quad \sum_{m=-s_1}^{s_2} p_{ij}(m)z^{-m} \right] \begin{bmatrix} r_i \\ r_j \end{bmatrix} \end{aligned} \quad (14)$$

where p_{ij} is the parameter vector that contains the coefficients of the F_{ij} feedforward controller. The process input u_{fi} can then be expressed as

$$u_{fi} = \Pi \begin{bmatrix} p_{ii} \\ p_{ij} \end{bmatrix} = \begin{bmatrix} r_i z^{s_1} & \dots & r_i z^{-s_2} & r_j z^{s_1} & \dots & r_j z^{-s_2} \end{bmatrix} \begin{bmatrix} p_{ii} \\ p_{ij} \end{bmatrix} \quad (15)$$

where $r_i z^{-m} \in R^{n \times 1}$ denotes the reference trajectory r_i with a delay of m sampling times. Through (13) and (15), gradient $\nabla e(p)$ can be deduced as

$$\nabla e(p) = \begin{bmatrix} T_{xx} & T_{xy} \\ T_{yx} & T_{yy} \end{bmatrix} \begin{bmatrix} \Pi & 0 \\ 0 & \Pi \end{bmatrix} = T_{\text{toep}} \Upsilon \quad (16)$$

where T_{toep} denotes the integrated process sensitivity Toeplitz matrix, and Υ denotes the integrated shift reference trajectory matrix. To obtain gradient $\nabla e(p)$, both model-based and data-driven approaches are enabling methods to solve matrix T_{toep} . Specifically, the model-based approach obtains the Toeplitz matrix T_{ij} by approximating the state-space dynamic model, which is derived from identification, whereas the data-driven approach obtains the gradient through the impulse response experiment. As the high-order dynamics and inherent model uncertainty exist in most practical applications, T_{toep} is obtained by the data-driven approach in this paper. Noting that gradient $\nabla e(p)$ is determined by r and T_{toep} , those are both uncorrelated with the servo error, and the estimation of gradient $\nabla J(p)$ is unbiased [27]. Furthermore, as no redundant experiment is needed to obtain gradient $\nabla e(p)$ in each iteration, the design procedure is comparatively simplified.

D. Hessian Inverse Problem

Existing researches indicate that the nonminimum phase characteristic is a main reason to enlarge the condition number of the Toeplitz matrix, which may acquire unrestricted inversion-based feedforward signal [28]. As the update calculation of (5) involves inverting the Hessian matrix, which is derived by the Toeplitz-based gradient, the bound of the feedforward signal should be restricted.

In numerical calculation, a large condition number of the Hessian matrix will undesirably raise the high-frequency magnitude of the feedforward signal, which may oscillate the control system. Furthermore, with the increasing order of the decoupling feedforward controller, the Hessian inverse problem will become severe as the dimension of the Hessian is multiplied. To tackle the inverse problem, the stable inversion method is employed to decrease the high-frequency energy.

Combining (7) and (16), the Hessian matrix is expressed as

$$R = 2(T_{\text{toep}} \Upsilon)^T \lambda (T_{\text{toep}} \Upsilon). \quad (17)$$

The stable inversion is achieved by the Moore–Penrose generalized inverse method. Considering the singular value decomposition of the Hessian as

$$R = [V_1 | V_2] \begin{bmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} \quad (18)$$

where Σ_2 contains the relatively small singular values, and Σ_1 contains others. Matrices U_i and V_i contain the singular vectors. To reduce the influence of the small singular values Σ_2 in the Hessian inversion and then restrict the bound of the updated feedforward signal, the small singular values Σ_2 is omitted. By setting the small singular values Σ_2 as zero, the inversion problem can be simplified as a least squares

approximation. Using the Moore–Penrose generalized inverse method, the inversion matrix can be computed as

$$R^{-1} \approx U_1 \Sigma_1^{-1} V_1^T. \quad (19)$$

Thus, as the approximation can effectively restrict the bound of the high-frequency energy of the controller, this inversion method is suitable to solve the problem that the updated feedforward signal oscillates the feedback loop. The performance of the method will be displayed by simulation in Section II-E.

With a stable feedback controller, the overall control system is asymptotically stable if the feedforward signal is bounded and convergent as the iterative number goes to infinity [16]. Thus, the asymptotical stability of the system is determined by the convergence characteristic of the designed feedforward control signal, which is discussed in the Appendix. To sum up, the design procedure of the proposed data-driven decoupling feedforward control can be described as follows.

- 1) Obtain T_{toep} by the impulse response experiment.
- 2) Use (16) and (17) to calculate $\nabla e(p)$ and the Hessian matrix, respectively.
- 3) Execute the singular value decomposition of the Hessian matrix and justify whether the Hessian matrix has a relatively small singular value. If the inverse problem appears, employ (19) to approximate the Hessian inversion.
- 4) Set the feedforward controller parameters to be zero.
- 5) Execute a motion task to obtain the servo error signal.
- 6) Use (5) to update all the controller parameters.
- 7) Return to step 5) and the parameters will converge to the optimal value iteratively.

E. Simulation Example

To illustrate the performance of the method, a 2×2 MIMO system is considered in the MATLAB simulation. A typical continuous-domain transfer function [12] is chosen as

$$P(s) = \begin{bmatrix} \frac{1.6 \times 10^5}{s^2(s^2 + 126s + 1.6 \times 10^6)} & \frac{s^2 + 20s + 2.5 \times 10^5}{5 \times 10^8 s^2} \\ \frac{s^2 + 15s + 1.4 \times 10^5}{1.4 \times 10^8 s^2} & \frac{3.6 \times 10^5}{s^2(s^2 + 377s + 3.6 \times 10^6)} \end{bmatrix} \quad (20)$$

which possesses a double integrator and a resonance characteristics. With a standard proportional–integral–derivative (PID) controller employed to guarantee the robust stability of the feedback loop, the proposed decoupling scheme is applied to the motion system. In the simulation, the x -direction tracks the trajectory, which is later described in Section III-C, whereas the y -direction keeps positioning.

First, the value of the iterative parameters should be determined. The weight matrix λ in (3) is set to be an identity matrix since both directions need to achieve the same level of accuracy. According to the characteristics of the inverse plant dynamics shown in Fig. 3, the order 4 and 6 offer a good compromise to F_{xx} and F_{xy} , respectively. T_{toep} is obtained by the impulse response simulation to generate the gradient from (16). With these initial parameters defined, the iterative simulation starts with the zeroed parameter vector p in (5).

Fig. 3 shows the final accuracy of the proposed method in the time domain and the dynamics comparison in the frequency domain, and “before” and “after” mean the performance with the initial and the updated parameter vector p , respectively.

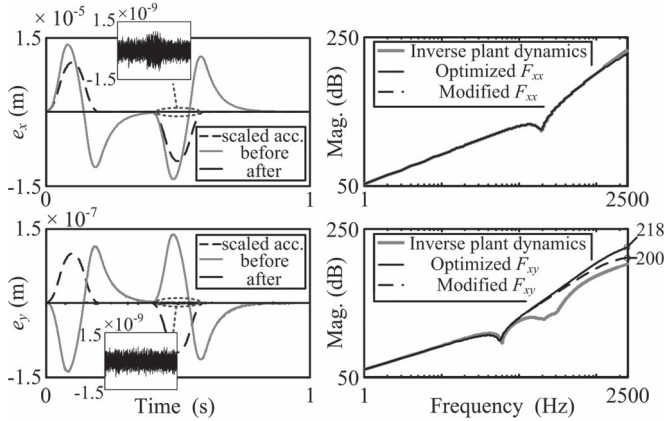


Fig. 3. Simulation results. (Left) Tracking error comparison between (gray) before and (black) after optimization. (Right) Optimized controller and the inverse plant dynamics. Mag. denotes magnitude. Scaled acc. means acceleration is proportionally changed.

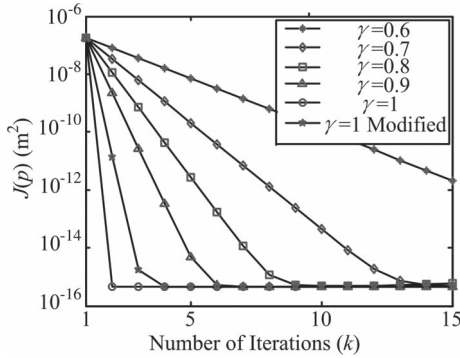


Fig. 4. Values of the objective function with different γ .

In the time domain, the method almost completely compensates the effect of the crosstalk since the positioning error of the y -direction is reduced to the same level of system noise. In the frequency domain, the feedforward controller can fit the inverse plant dynamics well in low frequency. However, the high-frequency dynamics of the controller F_{xy} is amplified. Fig. 3 shows that the “modified” controller has a lower high-frequency magnitude than that of the “optimized” controller, in which the “modified” controller uses the stable inversion method to avoid the oscillating problem while the “optimized” controller does not.

In addition, the convergence property with different step sizes of γ is discussed and is shown in Fig. 4. It indicates that the optimal point can be arrived when $\gamma = 1$ by only one trial, and the corresponding proof is included in the Appendix. It can also be seen that the modified controller will reduce the convergence rate but not affect the convergence accuracy.

III. APPLICATION TO AN ULTRAPRECISION MOTION STAGE

Here, the data-driven decoupling scheme is applied to an ultraprecision motion stage, which needs to achieve high-velocity and high-precision motion performance. Nowadays, the dual-stroke stage, which consists of a short-stroke (SS) stage for fine motion and a long-stroke (LS) stage for coarse motion, is widely used to achieve these criteria [29]–[31].

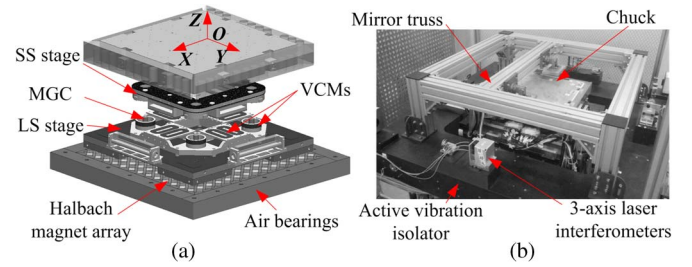


Fig. 5. (a) Schematic and (b) real assembly of the dual-stroke stage.

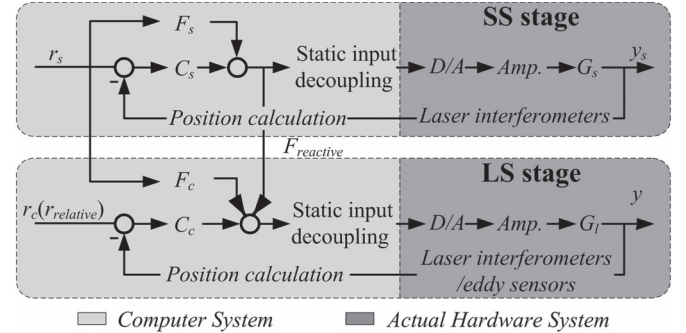


Fig. 6. Control scheme of the dual-stroke motion stage.

A. Structure of the System

Fig. 5 shows the structural configuration of the dual-stroke stage developed in our laboratory. The SS stage is driven by voice coil motors to achieve contactless 6-DOF movement. Considering heat dissipation, magnetic gravity compensators are mounted to counteract the gravity. The displacement of the SS stage is measured by laser interferometers. The LS stage is a friction-free X – Y planar actuator. The slider includes the three-phase coils, and the stator employs a Halbach permanent-magnet array. The stage is controlled in horizontal DOFs measured by eddy sensors and laser interferometers, whereas the vertical DOFs are restricted by air bearings.

B. Dynamics and Motion Control

Fig. 6 shows the concept of the control system. The SS loop is measured by laser interferometers and is controlled in absolute coordinate system. The LS loop is synchronized with the SS stage, which has two control modes: One is to keep a constant relative displacement with the SS stage, and the other is to track the same trajectory of the SS stage. The LS stage is measured by eddy sensors under the first mode and is measured by laser interferometers under the second mode. Combining with the computer-aided model, static input decoupling is employed to achieve system diagonal dominance.

The proposed scheme is applied on the SS stage in this paper, since similar performance can be obtained on the LS stage. First, six SISO feedback controllers, consisting of a standard PID controller, several notch filters, and a low-pass filter, are used to achieve fine system robust stability. The horizontal dynamics of the 6-DOF stage is shown in Fig. 7, and the other three DOFs have similar dynamic characteristics. The magnitudes of the off-diagonal terms show that the influence of the crosstalk among internal systems is relatively remarkable.

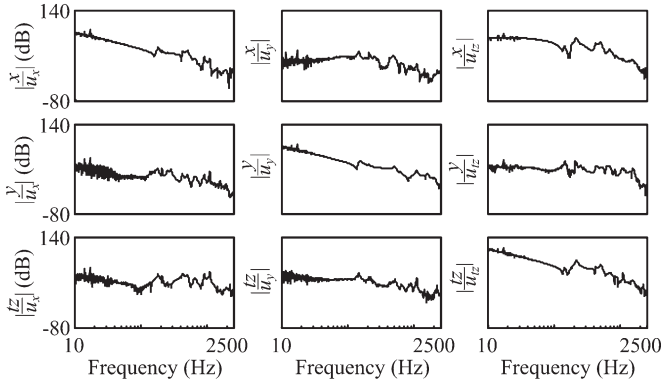


Fig. 7. Horizontal dynamics of the 6-DOF SS stage. The symbol $|u_i/u_j|$ denotes the ratio between the output of the i -direction and the input of the j -direction.

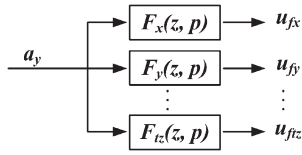


Fig. 8. Block diagram of the MIMO feedforward structure.

To achieve the required tracking accuracy, the crosstalk cannot be ignored. As the dynamics show the high-order and uncertain features, model-based decoupling methods might be difficult in achieving excellent performance.

C. Gradient Approximation

As the y -direction of the stage is the working direction, y -DOF tracks the trajectory, and other DOFs keep positioning in the experiment. Thus, the controller structure can be reshaped, as shown in Fig. 8. The feedforward controller parameter vector is then defined as $p = [p_{xy}^T p_{yy}^T p_{zy}^T p_{txy}^T p_{tyy}^T p_{tzy}^T]^T$.

The gradient of the tracking error $\nabla e(p)$ is derived as

$$\nabla e(p) = \begin{bmatrix} T_{xx} & \cdots & T_{xz} \\ \vdots & \ddots & \vdots \\ T_{zx} & \cdots & T_{zz} \end{bmatrix}_{6n \times 6n} \begin{bmatrix} \bar{\Pi} & 0 \\ & \ddots \\ 0 & \bar{\Pi} \end{bmatrix} = \bar{T}_{\text{toep}} \bar{\Upsilon} \quad (21)$$

where $\bar{\Pi} = [r_y z^{s_1} \cdots r_y z^1 \ r_y \ r_y z^{-1} \cdots r_y z^{s_2}]$.

To obtain the gradient, the integrated shift reference trajectory matrix $\bar{\Upsilon}$ and the integrated process sensitivity Toeplitz matrix $\bar{T}_{\text{toep}}$ should be obtained first. The matrix $\bar{\Upsilon}$ is determined by the y -direction reference trajectory, and the matrix $\bar{T}_{\text{toep}}$ is obtained by the data-driven approach.

1) Reference Trajectory: To smooth the control force, the high-order trajectory planning is a useful method. The fourth-order reference trajectories are used in the experiment, as shown in Fig. 9.

2) Integrated Process Sensitivity Toeplitz Matrix: The 6-DOF output time series is recorded when injecting a unit impulse at each freedom process input point, respectively. The average of ten times impulse responses in the experiments is shown in Fig. 10. Only the horizontal results are depicted since the other 3 DOFs have similar characteristics.

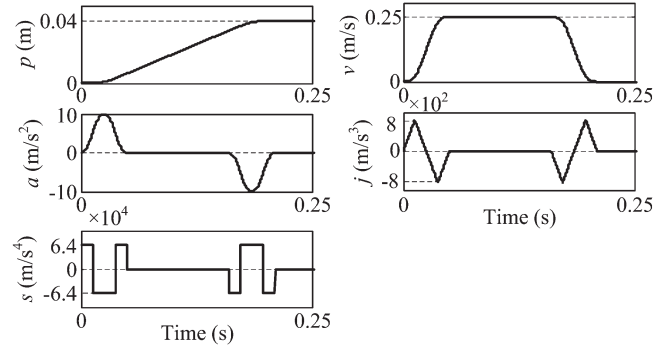


Fig. 9. Fourth-order reference trajectories. Herein, p , v , a , j , and s denote position, velocity, acceleration, jerk, and snap, respectively.

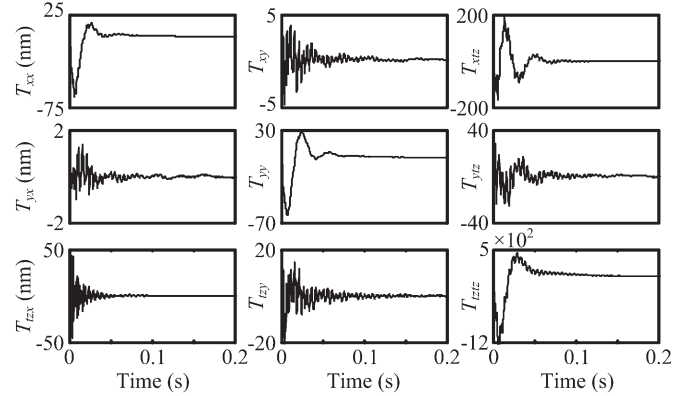


Fig. 10. Result of the horizontal impulse response experiment.

D. Performance Evaluations

To begin with, the inertia feedforward controller based on the dynamic model is employed. However, the behavior of the crosstalk among the axes still does not reach satisfactory. Thus, the proposed scheme is subsequently applied to further attenuate the effect of the crosstalk.

Noting that the system has a delay of four sampling times, which can be tested by the impulse response experiment, the ahead order s_1 is chosen as 4. Moreover, as the number of the main resonance peaks is roughly equal to 3 and each resonance could be simplified as a 2-order model, the lagging order $s_2 = 6$ offers a good compromise. The diagonal weight matrix λ is equal to $\text{diag}(1, 1, 0.5, 0.01, 0.01, 0.1)$ since the desired tracking accuracy of the translational DOFs is more precise than the rotational DOFs. The step size γ_k is set as 1, and the initial parameter vector is zero. As the constant velocity interval is the performance-relevant interval, the optimization interval is chosen to be $[0.02, 0.16]$ s, which contains partial interval of the acceleration and all interval of the constant velocity.

1) Time-Domain Behavior: Fig. 11 shows the tracking error before and after decoupling. It can be observed that the error of each DOF is attenuated, particularly in the y - and tx -directions. The values of the objective function before and after optimization are $8.75 \times 10^{-8} \text{ m}^2$ and $9.21 \times 10^{-10} \text{ m}^2$, which validate the improvement of the tracking performance. However, it can be also seen that only small improvement exists in the ty - and tz -directions. The reason is that the FIR structure has more capacity to decrease the low-frequency energy than

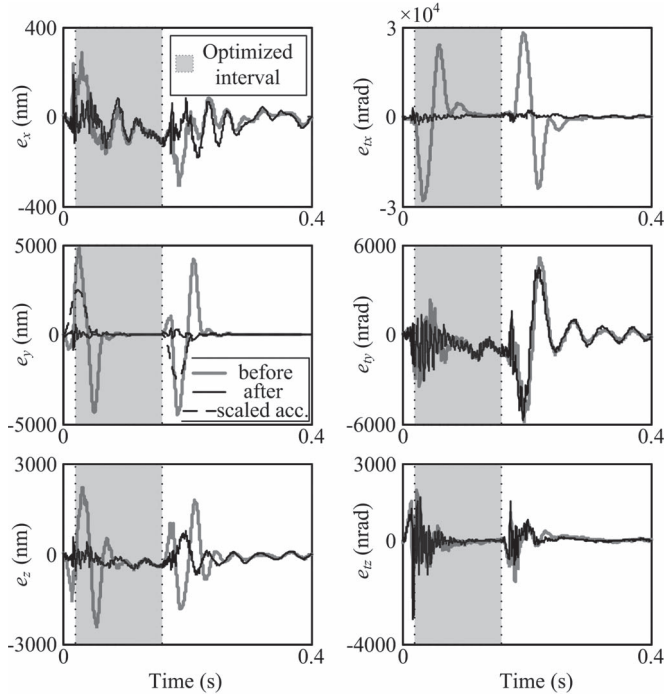


Fig. 11. Tracking error (gray) before and (black) after optimization.

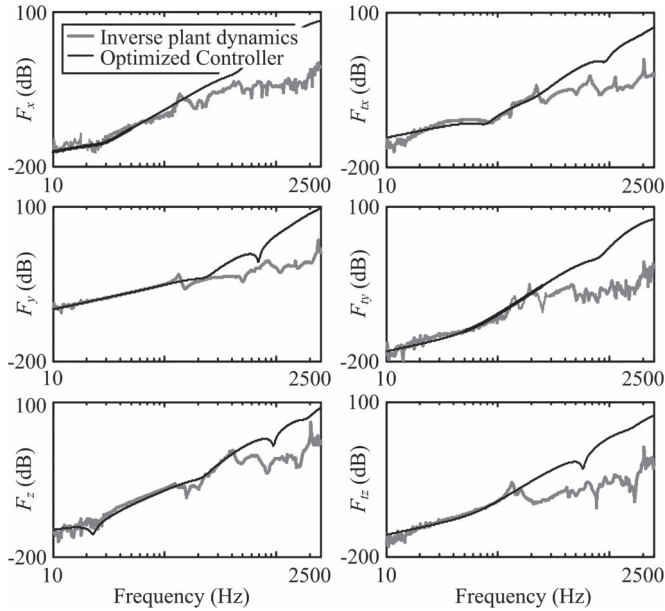


Fig. 12. Bode diagrams of the inverse dynamics and the optimized controller.

the high-frequency energy, since the errors of the optimization interval in the two directions appear in the high-frequency oscillation characteristics.

2) Frequency-Domain Behavior: Fig. 12 shows the bode diagrams of the inverse plant dynamics and the optimized feedforward controller. It can be seen that the optimization algorithm is suitable for approximating the low-frequency inverse dynamics. However, in high frequency, it does not fit well. There are several main reasons: 1) The stage has a lot of resonances distributing over 100 Hz that affect the fitting performance since high-order dynamic characteristics cannot be completely compensated by the controller; 2) the perfor-

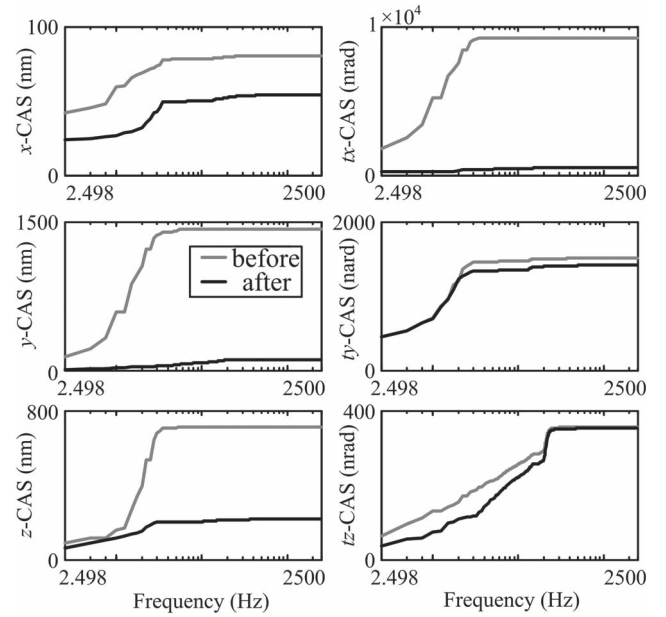


Fig. 13. CAS of servo error (gray) before and (black) after optimization.

mance of the FIR structure is restricted in high frequency since the energy of the servo error mainly locates in low frequency; and 3) model uncertainty and modeling error are inevitable, which renders the mismatch between the nominal and the actual model, particularly the off-diagonal terms. In addition, it is not necessary for the data-driven controller to focus on the inverse plant characteristic. Instead, only the servo error induced by the reference trajectory should be taken into account.

Fig. 13 shows the cumulative amplitude spectrum (CAS) of the tracking error before and after optimization. The CAS presents the energy distribution of the servo error in the frequency domain, and the final value of the CAS is equal to the standard deviation of the servo error [32]. Compared with the original error, the proposed algorithm can significantly diminish all the 6-DOF servo errors below 300 Hz, demonstrating effectiveness in practical applications.

IV. CONCLUSION

This paper has developed a data-driven iterative decoupling feedforward control scheme to address the ubiquitous crosstalk problem for practical MIMO motion control systems. Compared with model-based decoupling methods, the proposed control scheme utilizes the measurement data without demand for dynamic modeling. The performance of the control scheme has been verified through simulation, and the results of the convergence rate and accuracy have shown that the proposed method can rapidly attenuate the effect of the crosstalk. The proposed scheme has been also applied to an ultraprecision dual-stroke motion stage. With only one measurement data collection, the decoupling control scheme has obviously reduced the effect of the crosstalk since the value of the crosstalk criterion has been decreased from $8.75 \times 10^{-8} \text{ m}^2$ to $9.21 \times 10^{-10} \text{ m}^2$. In the frequency domain, this method can obviously compensate the low-frequency part of the servo error. For further servo performance improvement, the high-frequency dynamics compensation is an

important method in which the frequency-dependent criterion is a meaningful research subject. Overall, this paper has proposed an effective data-driven iterative decoupling feedforward control method, which can be applied to industrial motion control systems.

APPENDIX

The convergence of the proposed feedforward control scheme is discussed here. The second-order Taylor expansion of the objective function $J(p)$ at p_d is

$$J(p) = J(p_d) + \nabla J(p_d)^T \tilde{\alpha} + \frac{1}{2} \tilde{\alpha}^T \nabla^2 J(p_d) \tilde{\alpha} + O(\|\tilde{\alpha}\|_2^3) \quad (\text{A-1})$$

where $\tilde{\alpha} = p - p_d$ is the residual vector. As $\nabla J(p_d) = 0$ since p_d is the minimum, gradient $\nabla J(p)$ can be expressed as

$$\nabla J(p) \approx \nabla^2 J(p_d) \tilde{\alpha}. \quad (\text{A-2})$$

Noting (5), the iterative function can be rewritten as

$$p^{k+1} = p^k - \gamma^k R^{-1} \nabla^2 J(p_d) \tilde{\alpha}^k. \quad (\text{A-3})$$

As the Gauss–Newton direction has been chosen, R is equal to the Hessian $\nabla^2 J(p_k)$. Based on the approximation presented in Section II-C, the Hessian is unique, whereas the trajectory and the matrix T_{toep} are fixed. Thus, the iterative law of the residual vector in the vicinity of point p_d can be expressed as

$$\tilde{\alpha}^{k+1} = (1 - \gamma^k) \tilde{\alpha}^k. \quad (\text{A-4})$$

Thus, with $0 < \gamma < 2$ satisfying $\|\tilde{\alpha}^{k+1}\|_2 / \|\tilde{\alpha}^k\|_2 < 1$, the convergence of the iterative scheme can be guaranteed. It also indicates that the residual vector can converge to zero by only one iterative trial when the positive real scalar $\gamma = 1$.

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